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1

```
import java.sql.*;
```

```
import java.util.*;
```

```
public class MavenProject1 {
```

```
    static final String URL = "jdbc:mysql://localhost:  
3306/student_db";
```

```
    static final String USER = "root";
```

```
    static final String PASS = "";
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        HashMap<Integer, Integer[]> studentMarksMap =
```

```
        new HashMap<>();
```

```
    try {
```

```
        System.out.println("Enter Student ID: ");
```

```
        int id = sc.nextInt();
```

```
        sc.nextLine();
```

```
        System.out.println("Enter Student Name: ");
```

```
        String name = sc.nextLine();
```

2

```
System.out.println("Enter marks of 3 subject  
:");
```

```
Integer[] marks = new Integer[3];
```

```
int total = 0;
```

```
for (int i = 0; i < 3; i++) {
```

```
    System.out.println("sub subject " + (i+1) + ": ");
```

```
    marks[i] = sc.nextLine();
```

```
    total += marks[i];
```

```
}
```

```
studentMarksMap.put(id, marks);
```

```
String grade = calculateGrade(total/3);
```

```
insertData(id, name, total, grade);
```

```
displayRecords();
```

```
}
```

```
catch (Exception e)
```

```
    System.out.println("Error in Input: " + e.get  
        Message());
```

```
}
```

```
}
```

```
public static String calculateGrade (int avg) {
```

```
    if (avg >= 80)
```

```
        return "A+";
```

```
    else if (avg >= 70)
```

```
        return "A";
```

```
    else if (avg >= 60)
```

```
        return "A-";
```

```
    else if (avg >= 50)
```

```
        return "B";
```

```
    else
```

```
        return "F";
```

```
public static void insertData (int id, String  
    name, int total, String grade) {
```

```
    try (Connection conn = DriverManager.get  
        connection (URL, USER, PASS)) {
```

```
        String sql = "INSERT INTO students (
```

```
id, name, total, grade) VALUES (?, ?, ?, ?)";
```

```
PreparedStatement pstmt = conn.prepareStatement(sql);
```

```
pstmt.setInt(1, id);
```

```
pstmt.setString(2, name);
```

```
pstmt.setInt(3, totalgrade);
```

```
pstmt.setString(4, grade);
```

```
pstmt.executeUpdate();
```

```
System.out.println("In Data insert into  
database successfully!");
```

```
} catch (SQLException e) {
```

```
System.out.println("Database Error: "
```

```
+ e.getMessage());
```

```
}
```

```
}
```

```
}
```